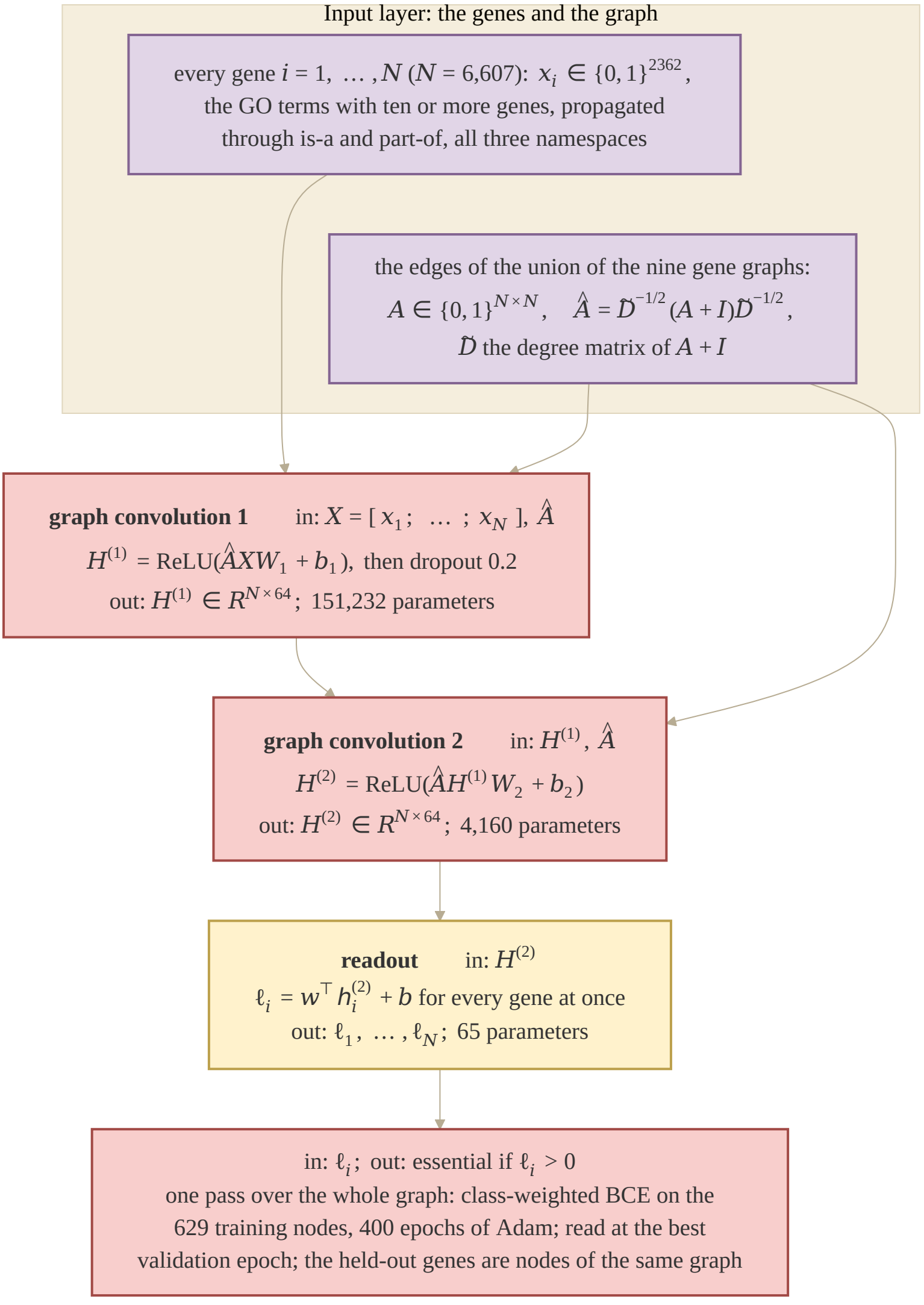

Input layer: the genes and the graph
every gene $i = 1, \ldots, N$ (N = 6,607): $x_i \in \{0,1\}^{2362}$,
the GO terms with ten or more genes, propagated
through is-a and part-of, all three namespaces
the edges of the union of the nine gene graphs:
$A \in \{0,1\}^{N \times N}$, $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$,
$\tilde{D}$ the degree matrix of $A + I$
graph convolution 1 in: $X = [x_1; \ldots; x_N]$, $\hat{A}$
$H^{(1)} = \mathrm{ReLU}(\hat{A}XW_1 + b_1)$, then dropout 0.2
out: $H^{(1)} \in R^{N \times 64}$; 151,232 parameters
graph convolution 2 in: $H^{(1)}$, $\hat{A}$
$H^{(2)} = \mathrm{ReLU}(\hat{A}H^{(1)}W_2 + b_2)$
out: $H^{(2)} \in R^{N \times 64}$; 4,160 parameters
readout in: $H^{(2)}$
$\ell_i = w^\top h_i^{(2)} + b$ for every gene at once
out: $\ell_1, \ldots, \ell_N$; 65 parameters
in: ℓ_i; out: essential if $\ell_i > 0$
one pass over the whole graph: class-weighted BCE on the
629 training nodes, 400 epochs of Adam; read at the best
validation epoch; the held-out genes are nodes of the same graph